

Air Quality Sensor Network Data Acquisition, Cleaning, Visualization, and Analytics: A Real-world IoT Use Case

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ABSTRACT

Monitoring and analyzing air quality is of primary importance to encourage more sustainable lifestyles and plan corrective actions. This paper presents the design and end-to-end implementation¹ of a real-world urban air quality data collection and analytics use case which is a part of the TRAFair (Understanding Traffic Flows to Improve Air Quality) European project [1, 2]. This implementation is related to the project work done in Modena city, Italy, starting from distributed low-cost multi-sensor IoT devices installation, LoRa network setup, data collection at LoRa server database, ML-based anomaly measurement detection plus cleaning, sensor calibration, central control and visualization using designed SenseBoard [3].

KEYWORDS

IoT Analytics, Anomalies Detection, Real-world Applications.

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1 INTRODUCTION

Urban air pollution is a serious public health hazard in many EU cities. In this paper, we employ low-cost IoT devices to monitor air quality in multiple location points at an urban scale. Using low-cost sensors is the only way to be able to carry out hyper-local monitoring in a city. However, managing low-cost AQ sensors is challenging. They are less reliable than the Air Quality Monitoring (AQM) legal stations and require a complex calibration process to obtain pollutant concentration data from the raw observations in millivolt they measured. The deployed implementation allows building a historical air quality dataset that contains accurate (calibrated with great precision close to AQM stations) and reliable (resilient LoRa networks) concentrations of air pollutants. These data can be the foundation for European Environment Agency (EEA), World Health Organization (WHO), other e-government bodies to design ML algorithms for advanced air-quality analytics.

¹Code released at <https://github.com/bharathsudharsan/Air-Quality-IoT-Analytics>

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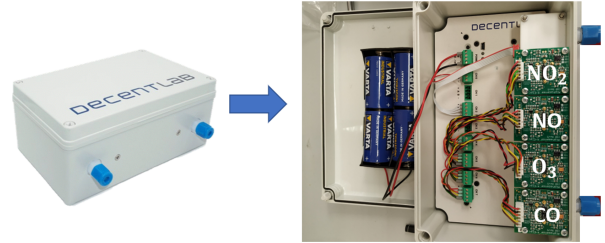
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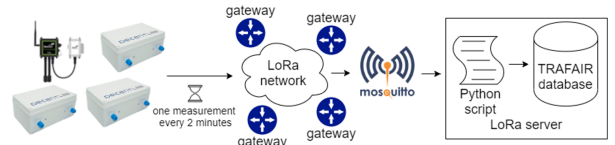
a. Multi-sensor LoRa based IoT device: Hardware view.



b. Installation locations of IoT devices: 12 points in Modena city, Italy.



c. Sensor data acquisition flow: From IoT devices to LoRa server.



d. SenseBoard architecture: Monitor, control, and visualization.

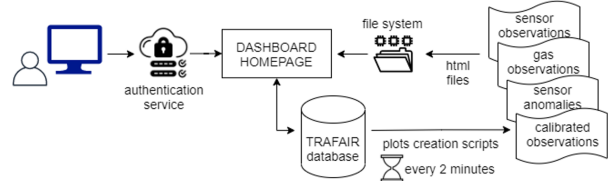


Figure 1: Infrastructure design and implementation for the air-quality dataset build and analytics use case.

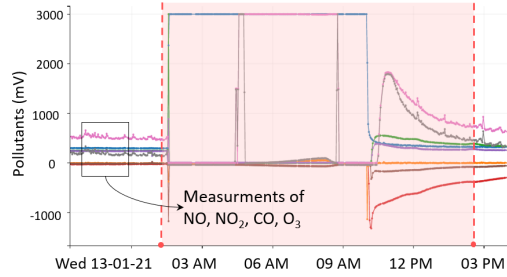
2 IMPLEMENTATION

In implementation *Step I* (Section 2.1), we explain the IoT device installation plus data acquisition process, then apply ML algorithms to detect anomalies and clean collected sensor data. In *Step II* (Section 2.2), we present the designed visualization platform to monitor and control the IoT devices and LoRa network. In *Step III* (Section 2.3), we present example analytics applications of our dataset.

2.1 Data Acquisition and Cleaning

Numerous IoT devices are exploited to monitor and collect the pollution level on an urban scale in Modena, Florence, Pisa, Livorno, Santiago de Compostela, and Zaragoza (cities in Italy and Spain). As shown in

a. Anomalous measurements due to device physical disturbance.



b. Anomalous measurements due to low battery voltage.

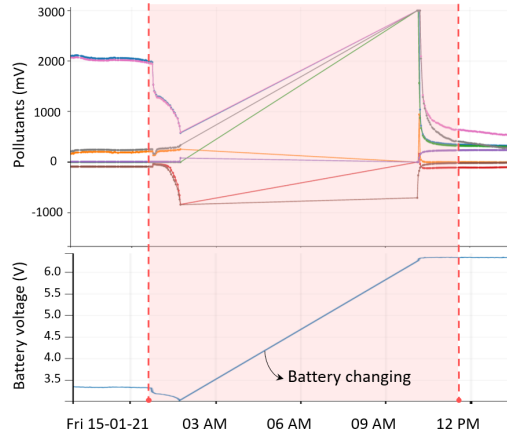


Figure 2: Frequent events that created anomalous sensor measurements during the air-quality dataset build.

Figure 1.b, 14 IoT devices are installed in 12 points of Modena, Italy (similar installation in remaining cities). We show the installed device’s hardware view in Figure 1.a, where each device contains 4 cells (sensors), one for each gas (NO , NO_2 , CO and O_x). Each cell measures the gas level through 2 channels (the auxiliary and the working channels) and provides a measure for each gas and channel in millivolts (mV). Sensor data acquisition flow is shown in Figure 1.c. Here, devices are connected to the LoRaWAN network through gateways and transmit data in a message every 2 minutes. When the gateways receive a message, they send it to the LoRa server where the MQTT (Message Queue Telemetry Transport) Broker Mosquitto is running. Then, the data are parsed in a python script and stored in a PostgreSQL database.

Air Quality Dataset Build Description. The data collection started in August 2019. The data rows generated by IoT devices contain NO , NO_2 , CO , O_x sensor measurements, temperature, humidity, and the battery voltage values with timestamps. At the time of writing, we have ≈ 20 months of thus built dataset, which we provide for researchers to perform air-quality analytics.²

Sensor Anomalies. As shown in Figure 2, the accuracy of raw measurements is affected by events, such as low battery voltage, weather conditions, air humidity, physical disturbances, and others. Since such events are prevalent in real-world distributed sensor deployment-based data collection practice, after data acquisition at the LoRa server, we need to centrally run algorithms to identify and remove such abnormal data patterns [4].

ML-based Anomalies Detection and Cleaning. In the project repository, we have run 12 unsupervised anomaly detection algorithms such as Angle-based Outlier Detection, Isolation Forest, clustering-Based Local

²Raw and calibrated data are available as open data on the open data portal of the Emilia Romagna region (<https://dati.emilia-romagna.it/>) and also available on the National and European data portals.

Outlier, and other algorithms on the built TRAFair air quality dataset. The anomaly score of each model type is provided along with the T-distributed Stochastic Neighbor Embedding (3D) and Uniform Manifold Approximation and Projection (2D) plots. We provide users the freedom to use any model of their choice to detect and remove anomalies (like the few shown in Figure 2). The output is clean data that can power advanced air quality analytics tasks.

2.2 SenseBoard Air Quality Data Visualization

The SenseBoard architecture [3] shown in Figure 1.d is our customized Python web app deployed on a High Performance Computing (HPC) Debian machine with 32 Intel(R) Xeon(R) Silver 4108 CPUs @ 1.80GHz and 256 GB RAM. Environmental experts can interact with the SenseBoard’s rich UI to obtain detailed visualization of live or historical (day, week, month) air quality data patterns.³ Each view is described in the following:

Sensor Observations. Produces 6 plots using the data collected from the user selected location/IoT device, that show the (i) relative humidity (%); (ii) temperature ($^{\circ}C$); (iii) battery (V); 4 gases for auxiliary and working channels (iv) as raw values (mV); (v) as calibrated through original factory calibration ($\mu g/m^3$); (vi) as calibrated through TRAFair calibration algorithms ($\mu g/m^3$).

Gas Observations. A plot for user-selected gas type that contains data collected from all existing IoT devices. This view can enable instant comparison of air quality patterns in selected locations.

Calibrated Observations. Displays calibrated data obtained by the application of ML algorithms to raw observations and certified air quality stations observations.

In addition to visualization, SenseBoard can also be used to control the status of the LoRaWAN sensor network, supervise calibration processes, update/add sensor locations, identify need for maintenance (battery change, physical damage repairs).

2.3 Air Quality Data Analytics

The battle against air pollution will be won with data. Following are few potential applications of the built dataset; (i) Train ML algorithms that can be centrally deployed (on a cloud platform) and give access to various APIs for forecasting/predicting (model inference) urban air quality in any given location point of Spain and Italy; (ii) Generate air quality index heat-map of past, present, future to rank the area’s emission levels and track changes; (iii) Assessment, intervention, and use for e-government decision support (since reliable historic data) when forming pollution reducing policies.

3 CONCLUSION

This paper briefly presented the TRAFair project deployed in Modena, Italy. We explained the end-to-end implementation of this real-world IoT use case from data acquisition until analytics. We believe the data and code that we made freely available to greatly aid research projects in the air quality analytics domain.

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- [4] Bharath Sudharsan, John G Breslin, and other. 2021. Porting and Execution of Anomalies Detection Models on Embedded Systems in IoT: Demo Abstract. In *International Conference on Internet-of-Things Design and Implementation (IoTDI)*.

³SenseBoard system demo:

https://www.youtube.com/watch?v=cfE7IJ0ZvNc&ab_channel=BigVisWorkshop